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MATLAB 4

Exercise

Consider the predator-prey system $\begin{cases} \frac{dR}{dt} = 2\left(1 - \frac{R}{3}\right)R - RF \\ \frac{dF}{dt} = -2F + 4RF \end{cases}$

a. Use solve to find all the equilibrium solutions.

Command Window Screenshot

Command Window

```
>> syms R F
>> S = solve(2 * (1 - R/3) * R - R * F == 0, -2 * F + 4 * R * F == 0, [R, F])

S =

  struct with fields:

    R: [3×1 sym]
    F: [3×1 sym]

>> S.R

ans =

  0
  3
1/2

>> S.F

ans =

  0
  0
5/3
```

% [3x1 sym] means that there are three (3) equilibrium points.

b. Plot the solution curve corresponding to the initial conditions $R(1) = 1$, *and* $F(1) =$ 3.7 over the interval $0 \leq t \leq 30$. For the axis, use [0, 2, 0, 6].

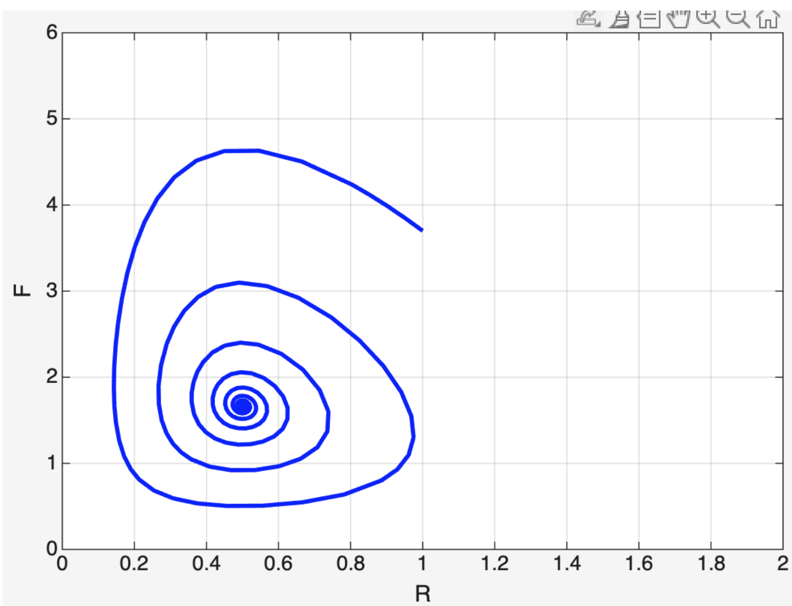
Command Window Screenshot

Command Window

```
>> f = @(t,y) [
    2*(1 - y(1)/3)*y(1) - y(1)*y(2);
   -2*y(2) + 4*y(1)*y(2)
];
>> timespan = [1 30];
>> y0 = [1; 3.7];
>> [t,y] = ode45(f, timespan, y0);
>> plot(y(:,1), y(:,2), 'b', 'LineWidth', 2)
>> xlabel('R')
>> ylabel('F')
>> axis([0 2 0 6])
>> grid on
>> hold on
>> axis tight
>> title('Solution Curve for R(1)=1, F(1)=3.7')
```

% Line 2 of the code represents dR/dt (rabbits, prey) and Line 3 represents dF/dt (foxes, predators)

Plot:



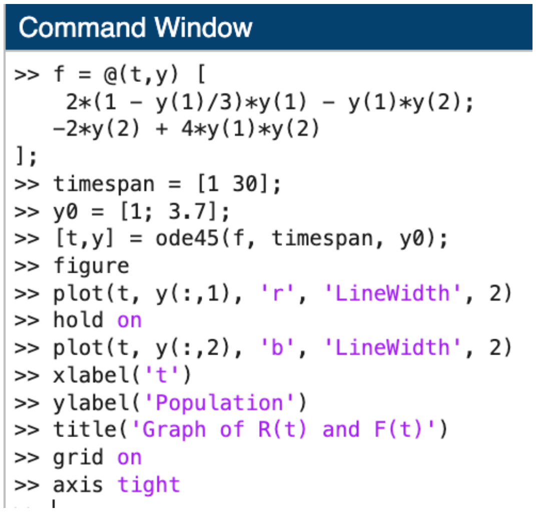
c. Describe the fate of the prey and predator populations based on the solution curve.

The solution curve gradually moves in a "spiral" shape towards the interior equilibrium point (R,F)=(1/2,5/3). As a result, both populations remain over time and neither goes extinct.

d. Display graphs of the solutions $R(t)$ and $F(t)$ corresponding to the initial conditions

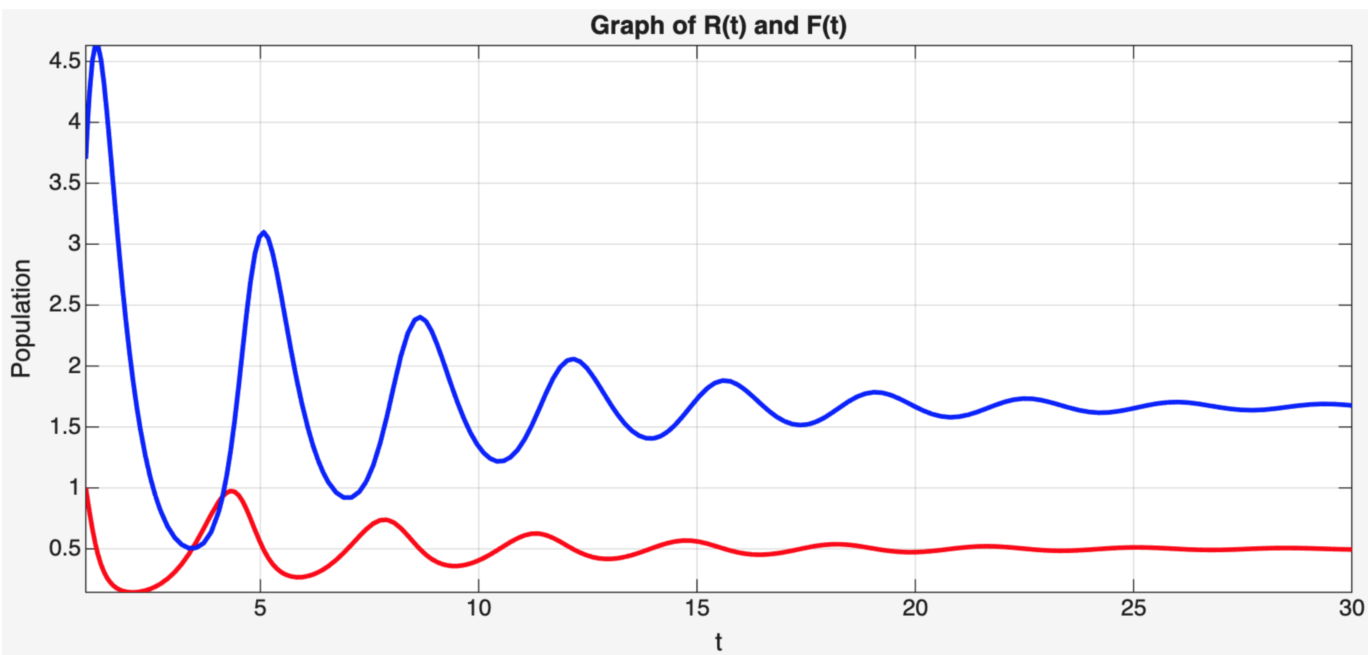
$R(1) = 1$, *and* $F(1) = 3.7$ over the interval $0 \le t \le 30$. Compare with part (c).

Command Window Screenshot



```
Command Window
>> f = @(t,y) [
    2*(1 - y(1)/3)*y(1) - y(1)*y(2);
   -2*y(2) + 4*y(1)*y(2)
];
>> timespan = [1 30];
>> y0 = [1; 3.7];
>> [t,y] = ode45(f, timespan, y0);
>> figure
>> plot(t, y(:,1), 'r', 'LineWidth', 2)
>> hold on
>> plot(t, y(:,2), 'b', 'LineWidth', 2)
>> xlabel('t')
>> ylabel('Population')
>> title('Graph of R(t) and F(t)')
>> grid on
>> axis tight
```

Graph:



% Comparison: The graph for R(t) and F(t) (rabbit and fox population vs time respectively) displays damped oscillations in both populations, with R(t) and F(t) converging to finite values. This aligns with the "spiral" trajectory observed in part c, confirming that the equilibrium is stable and that both species will coexist in the long term without either going extinct.